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Confidence-Aware Self-Prompting for Robust Polyp Segmentation Under Image and Domain Shifts

Phase 3 Report: Implementation & Execution

Group	F26-13
Track	A (Research & Development)
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Repository: github.com/kashishfatima999/Confidence-Aware-Self-Prompting-for-Robust-Polyp-Segmentation
Notebook: `F26-13.ipynb` (the executed Colab run behind every number in this report)

Summary. We trained a YOLOv8-m detector on our Kvasir-SEG split, reproduced the YOLO → SAM 2 baseline of Mansoori et al. [1], and implemented our method, Confidence-Aware Self-Prompting (CASP v1), together with four control experiments. CASP raises per-image Dice from 0.831 to 0.891 on Kvasir-SEG test and from 0.677 to 0.757 on CVC-ClinicDB, a dataset the detector never saw. The control runs show that the gain comes from keeping the boxes the detector is unsure about, not from the extra checking steps. Every number is read from the result files of the runs (`docs/tables/phase3/`); every choice is recorded in `docs/decision_log.md` (DL-30 to DL-38).

1. Model and algorithm selection

Detector: YOLOv8-m [2]. The baseline paper [1] uses it and reports that the medium size beat both the small and the large model, so it is the fair choice for a reproduction. It is also small enough (about 25M parameters) to train on a free Colab T4 in under half an hour.

Segmenter: SAM 2, large, frozen [3]. SAM 2 draws a precise mask from a single box prompt without any training on polyps. Our Phase 2 smoke test showed it reaching Dice 0.96–0.97 when the box is right and 0 when the detector misses, which told us that *the box, not the segmenter, is where a method can help*. Keeping SAM 2 frozen keeps the comparison with the baseline clean.

Our method: CASP v1. The baseline sends every box with YOLO confidence ≥ 0.5 to SAM 2 and throws away the rest, so a polyp the detector found but was only 40% sure of gets no mask at all. CASP uses two thresholds instead of one and treats each box by its confidence c:

Confidence	Group	What happens
$c \geq$ `t_high`	confident	The box goes to SAM 2 unchanged, as in the baseline.
`t_low` $\leq c <$ `t_high`	uncertain	(1) SAM 2 is prompted four ways: the box, the box grown by 10% and by 20% on every side, and the box plus a point at its centre; (2) the mask SAM 2 itself scores highest (its predicted IoU) is kept; (3) *self-prompting*: a tight box and the centre of that mask are given back to SAM 2, and the new mask is kept if it scores at least as well; (4) the mask is accepted only if SAM 2's score is ≥ `s_min` and it fills ≥ 10% of its box.
$c <$ `t_low`	dropped	Ignored.

Why this design. It needs no training: every step only re-prompts the frozen SAM 2, and the image is encoded once, so each extra prompt is cheap. Setting `t_low` = `t_high` = 0.5 without the acceptance rule gives exactly the baseline, so any difference between the two can only come from the new rules. We considered and rejected fine-tuning SAM on perturbed prompts and training a box-refinement network (both need training time we do not have and would change the frozen-SAM setting), and using Gaussian-YOLO localisation uncertainty (it requires changing the detector); see DL-34.

Control experiments (DL-35), so that we can see where any gain comes from: **T** lowers the baseline threshold to `t_low` and does nothing else; **C-all** checks every box, not only uncertain ones; **C-noiter** drops the self-prompting step; **O** (oracle) gives SAM 2 the ground-truth boxes, the best any detector could do.

2. Implementation details

- **Frameworks:** PyTorch 2.11 (CUDA), Ultralytics 8.4.174 for YOLOv8 training and inference, Meta's `sam2` package (pinned at commit `2b90b9f`) with the `sam2_hiera_large.pt` checkpoint; NumPy, Pillow, OpenCV and pandas for data and metrics; PyYAML for configs.
- **Hardware:** Google Colab, Tesla T4 (15 GB). Data preparation (Phase 2) runs on CPU.
- **Code** (`src/`): `train_yolo.py` (training recipe from `configs/yolo_train.yaml`), `pipeline/detect.py` (detection cache), `pipeline/sam2_prompter.py` (SAM 2 wrapper: encode once, prompt many), `pipeline/refine.py` (CASP), `pipeline/run_experiment.py` (one config + one dataset → results), `eval/metrics.py` and `eval/summarize.py` (metrics and tables). One YAML file per experiment in `configs/experiments/`.
- **Testing:** 65 unit tests (the CASP logic is tested with a mocked SAM 2) run without a GPU on every push through GitHub Actions. The Colab notebook only calls these scripts.
- **Metrics:** Dice and IoU per image, averaged, with a bootstrap 95% confidence interval; an image with no prediction counts as 0, so a missed polyp is never hidden. We also report the paper's *pooled* Dice (all pixels of a dataset at once), the number of *missed* images, and box precision and recall at IoU ≥ 0.5.

3. Code workflow: data input → model → output

Input
Processed images and masks (`data/processed/`): Kvasir val / test, ClinicDB, 9 corrupted sets

→

Detector
YOLOv8-m once per dataset at conf 0.05; every box and score cached in `runs/detections/*.json`

→

CASP gate
Per box: confident / uncertain / dropped, by `t_low`, `t_high` (`refine.py`)

→

SAM 2
Box or candidate prompts, self-prompting, acceptance check; masks joined per image

→

Output
`per_image.csv`, `detections.jsonl`, `summary.json`, masks → results tables

The detection cache is the key design choice (DL-33): YOLO runs once, and every experiment only re-filters the saved boxes, so all six runs see exactly the same detector output and each run is one command, for example `python src/pipeline/run_experiment.py --config configs/experiments/casp.yaml --dataset kvasir_test`. Results are copied to Google Drive as soon as each run finishes, so a dropped Colab session costs one run, not a whole evening.

4. Training process

Data split (fixed in Phase 2): Kvasir-SEG [4], 700 train / 100 validation / 200 test images, seed 42. CVC-ClinicDB [5] (612 images, another hospital and endoscope) is used only for testing, never for training or tuning, and so is the corrupted copy of the Kvasir test split (blur, brightness/contrast and noise at three severities).

Detector training (DL-31). The recipe was written down before training and not changed afterwards:

Setting	Value	Reason
Start weights	COCO-pretrained YOLOv8-m	as in the paper
Epochs	50, no early stopping	baseline README
Image size	640 px	paper: 680, not a multiple of 32
Batch size	16	fits a T4 (paper: 64 on an A100)
Optimiser, augmentation	Ultralytics defaults	no tuning on our data
Seed	42, deterministic	repeatable run
Kept checkpoint	best validation mAP50-95	used by every experiment

Split	mAP50	mAP50-95	Prec.	Recall
Val (100)	0.953	0.750	0.960	0.869
Test (200)	0.943	0.747	0.867	0.908

Training took 25 minutes. The losses fall steadily and validation mAP50 first passes 0.9 at epoch 15 (Figure 1, left). mAP50 drops only one point from validation to test, so the detector did not overfit the validation images.

Choosing CASP's thresholds. SAM 2 is not trained, so the only parameters of our method are `t_low`, `t_high` and `s_min`. We ran all 27 combinations of `t_low` ∈ {0.15, 0.25, 0.35}, `t_high` ∈ {0.6, 0.7, 0.8} and `s_min` ∈ {0.6, 0.7, 0.8} on the **validation images only** and kept the highest mean Dice: `t_low` = 0.25, `t_high` = 0.6, `s_min` = 0.6, with validation Dice 0.8985 against 0.866 for the baseline on the same images. These values were written into the configs before any test run, and each test set was then run once per experiment. On validation, 0.15 was always worse than 0.25 or 0.35 (it lets in 18–19 false boxes instead of 10–12), and `s_min` = 0.7 and 0.8 gave identical results.

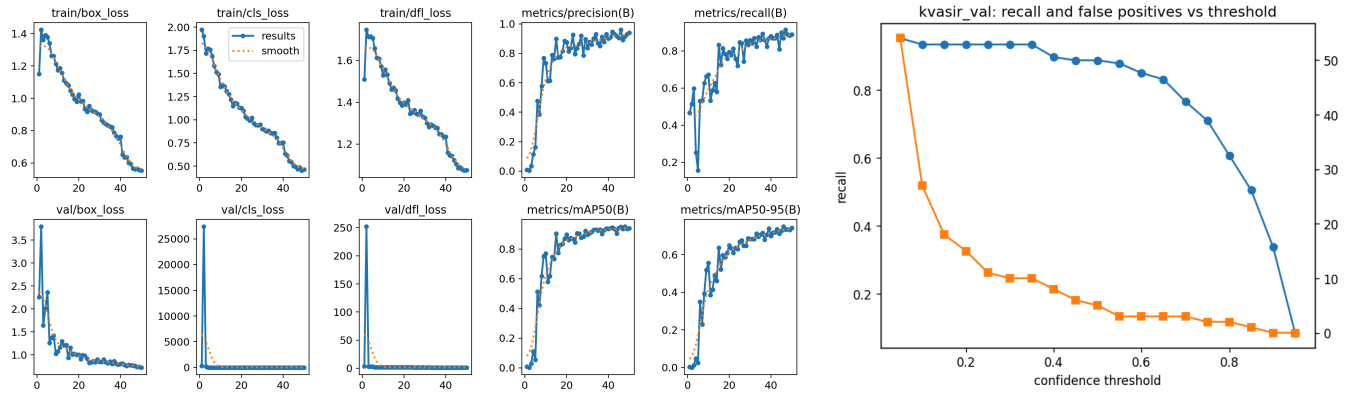


Figure 1. Left: YOLOv8-m training curves over 50 epochs. Right: on the validation split, polyp recall (blue) and false-positive boxes (orange) against the confidence threshold. From 0.5 down to 0.25, recall rises from 0.888 to 0.935 while false boxes only go from 5 to about a dozen; below 0.25 recall barely moves but false boxes shoot up. This is the room our method works in.

5. Initial results

Run	Kvasir-SEG test (200 images)				CVC-ClinicDB (612 images, never trained on)			
	Dice [95% CI]	Pooled	Missed	Box P / R	Dice [95% CI]	Pooled	Missed	Box P / R
B baseline	0.831 [0.792, 0.871]	0.878	18	0.877 / 0.865	0.677 [0.648, 0.710]	0.755	144	0.890 / 0.686
T threshold only	0.893 [0.871, 0.913]	0.901	1	0.807 / 0.947	0.759 [0.736, 0.785]	0.799	75	0.820 / 0.780
C CASP v1	**0.891 [0.868, 0.912]**	**0.894**	**1**	**0.802 / 0.942**	**0.757 [0.734, 0.783]**	**0.795**	**75**	**0.815 / 0.775**
C-all	0.885 [0.860, 0.907]	0.879	1	0.802 / 0.942	0.757 [0.734, 0.784]	0.794	75	0.808 / 0.769
C-noiter	0.892 [0.869, 0.913]	0.895	1	0.807 / 0.947	0.757 [0.734, 0.783]	0.795	75	0.813 / 0.774
O oracle	0.952 [0.943, 0.958]	0.951	0	1.000 / 1.000	0.945 [0.942, 0.948]	0.945	0	1.000 / 1.000

Dice: per image, misses counted as 0. Pooled: the paper's pooled Dice. Box P / R: box precision and recall at IoU ≥ 0.5. IoU and timings for every run are in `docs/tables/phase3/results.csv`.

- **The baseline reproduces.** On the paper's own pooled metric our baseline scores 0.878 Dice / 0.782 IoU on Kvasir-SEG test against the published 0.866 / 0.764. On ClinicDB the paper trains on ClinicDB itself, so its 0.951 is not comparable with our cross-dataset test; there the baseline misses 144 of 612 images (24%) completely.
- **CASP beats the baseline** by +0.061 Dice on Kvasir test and +0.080 on ClinicDB (paired bootstrap 95% CIs [+0.032, +0.090] and [+0.060, +0.100], both above zero). Missed images fall from 18 to 1 and from 144 to 75. It is also above the baseline on all nine corrupted settings, by +0.02 to +0.08 Dice; the gain grows with blur severity, until the strongest noise level, where the detector proposes nothing at all.
- **The gain comes from keeping uncertain boxes.** T, which only lowers the threshold, gets the same improvement (+0.062 and +0.082). The checking steps change Dice by −0.002, self-prompting by less than 0.001, and the acceptance rule never rejected a box: SAM 2's own score was at least 0.78 for every uncertain box, true or false, because it rates how clean its mask is, not whether the region is a polyp. Checking confident boxes too (C-all) is slightly worse on Kvasir (−0.006) at 2.5 times the prompts.
- **What is left is the detector.** With ground-truth boxes SAM 2 reaches about 0.95 on both datasets, so it segments the unseen hospital's polyps as well as the training ones. The price of our recall gain is false alarms: box precision falls from 0.877 to 0.802 on Kvasir and from 0.890 to 0.815 on ClinicDB (Figure 2). Building a filter that removes them is the first Phase 4 task.

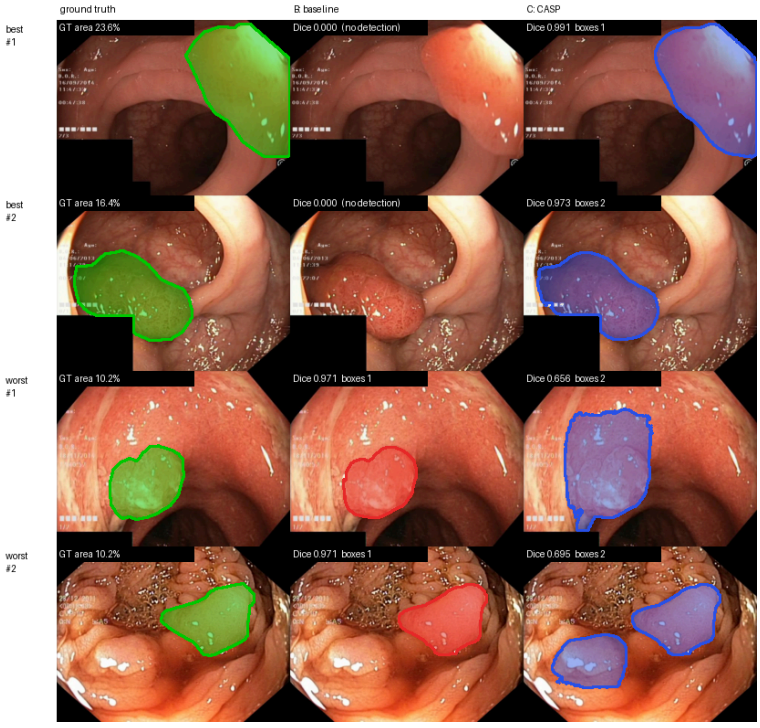


Figure 2. Kvasir test: ground truth (green), baseline B (red), CASP C (blue). Top two rows: the largest gains, both images the baseline missed completely. Bottom two rows: the largest losses, both caused by a second low-confidence box that C keeps and B did not.

6. Challenges faced during implementation

- **The baseline code [6] did not run as published** (notebook syntax in `.py` files, missing imports, a hard-coded threshold, and an evaluation that skips missed images). We re-implemented the pipeline and checked it against the paper on its own metric (§5).

- **Colab sessions dropped.** The free tier disconnected during a silent 50-minute run of the threshold grid and two hours of results were lost. The notebook now prints one line per run, saves each run to Drive as soon as it finishes and skips finished runs on restart. For long jobs we added a Kaggle version (`notebooks/05_kaggle_driver.ipynb`) that keeps running with the browser closed.
- **Package conflicts on Colab.** Installing our local requirements upgraded Pillow inside the running Colab kernel and broke the next import; on Colab we now install only the model packages and use Colab's own NumPy, Pillow and OpenCV.
- **Small validation set and one seed.** Neighbouring threshold settings differ by at most 0.006 Dice on 100 images, and the detector was trained once, so the confidence intervals cover image sampling only. The chosen thresholds still held on both test sets and all nine corruptions.
- **An acceptance rule that never fires.** We expected SAM 2's own score to reject wrong boxes; it did not (§5). We report this as it is (DL-38) and replace that signal in Phase 4.

Work division (full record in `project-division/Project-Division-Phase-3.md` §7). Kashish Fatima (lead): evaluation harness, first versions of the pipeline code, experiment configs, results tables and the Colab runs. Khushbakht Sohail: detector code (training recipe, detection cache), detector and baseline analysis. Zahra Saeed: SAM 2 wrapper, CASP code, experiment runner and notebooks, method and ablation analysis, Kaggle driver, this report.

References

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[3] N. Ravi *et al.*, "SAM 2: Segment anything in images and videos," 2024, *arXiv:2408.00714*.

[4] D. Jha *et al.*, "Kvasir-SEG: A segmented polyp dataset," in *MultiMedia Modeling (MMM 2020)*, Lecture Notes in Computer Science, vol. 11962. Cham, Switzerland: Springer, 2020, pp. 451–462.

[5] J. Bernal, F. J. Sánchez, G. Fernández-Esparrach, D. Gil, C. Rodríguez, and F. Vilariño, "WM-DOVA maps for accurate polyp highlighting in colonoscopy: Validation vs. saliency maps from physicians," *Comput. Med. Imag. Graph.*, vol. 43, pp. 99–111, Jul. 2015.

[6] "YOLO_SAM2: Official implementation of [1]," GitHub repository. [Online]. Available: https://github.com/sajjad-sh33/YOLO_SAM2 (accessed Oct. 2026).